Reliability of $V_{S,30}$ Evaluation from Surface-Wave Tests

C. Comina[1]; S. Foti[2]; D. Boiero[3]; and L. V. Socco[4]

Abstract: The reliability of surface-wave tests for the evaluation of $V_{S,30}$ in seismic site characterization is assessed with respect to both uncertainty and accuracy. The discussion of uncertainty is mainly focused on the implications of solution nonuniqueness in inverse problems; only the inversion uncertainty is considered within this work, omitting other possible issues such as nontrivial geological settings (e.g., lateral variations) or the influence of different processing procedures. A Monte Carlo approach has been used to select, through a statistical test, a set of shear-wave velocity models that can be considered equivalent with respect to fitting the experimental dispersion curve according to the information content (dispersion velocities and frequency range) and the experimental uncertainties. This set of equivalent solutions is then used to evaluate the uncertainty in the determination of $V_{S,30}$. Moreover, comparisons between the results obtained by surface-wave tests and invasive seismic methods are reported to assess the accuracy of $V_{S,30}$ evaluation by using surface-wave methods. It is shown that, given an adequate investigation depth, the solution nonuniqueness is not a major concern and that the results are comparable in most situations with the results of invasive tests providing an accurate estimate of $V_{S,30}$, even with simplified approaches. **DOI: 10.1061/(ASCE)GT.1943-5606.0000452.** *© 2011 American Society of Civil Engineers.*

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Introduction

The weighted average of shear-wave velocities in the shallowest 30 m ($V_{S,30}$) is often used for seismic site classification. $V_{S,30}$ is used, for example, in the National Earthquake Hazards Reduction Program (NEHRP) provisions [Building Seismic Safety Council (BSSC)1994] to separate sites into different seismic classes and a similar classification has been adopted also in European codes [e.g., European Committee for Standardization (CEN) 2004]. This classification is then used to determine the site-dependent seismic coefficients for earthquake-resistant design. Other uses of $V_{S,30}$ as a relevant seismic parameter include the generation of attenuation laws for predicting ground motion parameters in seismic hazard studies (Power et al. 2008).

Different geophysical techniques, either invasive (downhole, cross hole, P-S suspension logging) or noninvasive [surface-wave tests or horizontally polarized shear wave (SH) seismic refraction], can be used in the field to obtain $V_{S,30}$. It is important to emphasize that none of these methods provides "what can be deemed as an unbiased estimate" (Moss 2008) and that a correct evaluation of the uncertainty of the parameter is crucial. Invasive methods are generally considered more accurate, but they are not free of uncertainty.

Surface-wave tests are widely used to estimate $V_{S,30}$ because they are cost- and time-effective when compared to invasive methods, and they do not suffer from limitations related to site conditions that affect SH seismic refraction (e.g., presence of inverse layering or hidden layers).

Surface-wave tests are based on geometrical dispersion, which makes the velocity of propagation of Rayleigh waves frequency dependent in vertically heterogeneous media. The experimentally measured dispersion curve is used for the solution of an inverse problem aimed at the identification of soil model parameters (shear-wave velocities and thicknesses of a horizontally stratified medium). This paper does not attempt to provide guidelines for the selection of methods for obtaining reliable dispersion curves and for the solution of this inversion process. Rather, it is focused on evaluating the accuracy and uncertainty of these data if they are used for $V_{S,30}$ estimation.

Certainly a variety of approaches exists to determine experimentally a site-specific surface-wave dispersion curve. In active-source tests, waves are generated by using a seismic source (Stokoe et al. 1994; Park et al. 1999; Foti 2000), whereas passive-source tests are based on the analysis of microtremors (Horike 1985; Tokimatsu 1995; Louie 2001). Whereas in active-source tests it is usually very easy to generate and detect high frequency components, microtremors are typically rich in energy in the low frequency band. The resolvable frequency band affects the investigation depth (related to the maximum recorded wavelength) and the spatial resolution close to the ground surface (related to minimum recorded wavelength). A combination of both active and passive methods may be necessary to obtain a reliable evaluation of $V_{S,30}$ (depending on the strength of the active source). Combined use of passive and active methods has been suggested to improve both resolution and investigation depth (Tokimatsu 1995; Rix et al. 2002; Foti et al. 2007). Different studies have shown that the measurement uncertainty in active surface-wave data is typically below 5% in terms of coefficient of variation (CoV) of the Rayleigh wave phase velocity (Xia et al. 2002; Marosi

[1]Dept. of Earth Science (DST), Università degli Studi di Torino, via Valperga Caluso, 35 10125 Italy. E-mail: cesare.comina@unito.it

[2]Dept. of Structural and Geotechnical Engineering (DISTR), Politecnico di Torino, corso Duca degli Abruzzi, 24 10129 Italy (corresponding author). E-mail: sebastiano.foti@polito.it

[3]Dept. of Land Environment and Geoengineering (DITAG), Politecnico di Torino, corso Duca degli Abruzzi, 24 10129 Italy. E-mail: daniele.boiero@polito.it

[4]Dept. of Land Environment and Geoengineering (DITAG), Politecnico di Torino, corso Duca degli Abruzzi, 24 10129 Italy. E-mail: valentina.socco@polito.it

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and Hiltunen 2004a; Lai et al. 2005). These values are estimated from different acquisitions in the same testing configuration; hence, lateral variations in complex geological settings and errors induced by processing are not accounted for. Some attempts to propagate the measurements uncertainty on the estimated shear-wave velocity profile are also reported in the literature (Marosi and Hiltunen 2004b; Asten and Boore 2005; Lai et al. 2005).

Once the experimental dispersion curve is retrieved, different approaches can be used for the solution of the inverse problem aimed at the estimation of the shear-wave velocity profile. Most of them are based on deterministic gradient-based inversion techniques (e.g., Herrmann 1994; Lai and Rix 1998), so that the solution of the inverse problem is a single velocity profile. Because surface-wave inversion is strongly nonlinear, ill-posed, and mix-determined, the solution is nonunique, and several velocity profiles may comply with the experimental data. The final model chosen by deterministic methods is only one of the possible solutions, and the final result is very sensitive to the initial model. The inversion process can easily be biased by wrong choices in terms of model parameterization that lead the solution into local minima (Sambridge 2001; Luke et al. 2003; Wathelet et al. 2004). The nonuniqueness of the inverse problem solution represents one of the main causes of uncertainty in surface-wave analysis. In spite of this, it is often neglected, especially when deterministic inversion algorithms are used.

Global search methods avoid the linearization of the problem and mitigate the risk to converge in a local minimum. Moreover, they can provide a population of solutions from which it is possible to statistically asses the implications of solution nonuniqueness. In the present study, a Monte Carlo approach is used to evaluate the consequences of solution nonuniqueness, accounting also for measurement uncertainties. The procedure leads to the selection of a group of shear-wave velocity profiles, which can be considered equally good with respect to the available experimental information (i.e., the experimental dispersion curve and its associated measurement uncertainty). This approach leads to a consistent evaluation of inversion uncertainties in $V_{S,30}$ determination. A similar procedure has been adopted to study the influence of solution nonuniqueness in seismic-site-response studies (Foti et al. 2009).

In the following, a description of experimental testing and inversion procedures is reported. Then, a collection of several case histories in Italy is used to discuss the reliability and accuracy of surface-wave tests in $V_{S,30}$ evaluation. Finally, the experimental data set is also used to provide further validation of an existing approximate formula (Brown et al. 2000) for the evaluation of $V_{S,30}$.

Methods

Eight test sites have been considered in this study. In all of them, active-source surface-wave data have been collected, sometimes in combination with passive-source tests. Independent invasive seismic tests were also available as a benchmark at all sites. Details of the testing setups and acquisition parameters are reported in Table 1. For the active-source tests, linear arrays of 24 or 48 vertical geophones with 4.5 Hz natural frequency were used. The standard active source was a 5-kg sledgehammer; a weight-drop system (130 kg from 3 m) was used at Saluggia and Pontremoli sites; a self-propelled weight dropper has been used at Torre Pellice (drop height 1.6 m, total weight 750 kg). The distance from the source to the central point of the array are generally such that near field effects can be considered of minor influence; indeed the source-to-array center distance is typically larger that half of the maximum experimental wavelength (Rosenblad and Li 2011). Several independent acquisitions (at least 15, including also forward and reverse shots) were performed with the same acquisition setup. Dispersion curves were obtained from field data by using frequency-wave number (f-k) analysis (Gabriels et al. 1987; Foti 2000) implemented in the code Surface Wave Analysis Tool (SWAT), developed in the Matlab environment at Politecnico di Torino. For the passive-source tests, circular arrays of 50–75 m diameter with 8–24, 2 Hz vertical geophones (evenly spaced along the circumference) were used. Also in this case, several independent recordings were acquired. Dispersion curves were obtained from microtremors by using the frequency domain beamformer (FDBF) technique implemented in a Matlab code developed by Zywicki (1999). Compared to linear arrays used with the refraction microtremor (ReMi) method (Louie 2001), the use of two-dimensional (2D) arrays provides a more reliable estimate of phase velocities because it takes into account the actual direction of propagation of the microtremors, which is a priori unknown. This aspect is crucial, especially when the sources of microtremors are not homogeneously distributed around the testing site.

Table 1. Acquisition Parameters for the Case Histories

	Number of active channels	Geophone spacing (m)	Sampling rate (ms)	Time window (s)	Array length (m)	Source-array distance (m)
Active surface-wave tests						
Pontremoli	24	1 and 2	2	4.096	24 and 72	4
Saluggia	24	1 and 3	1	0.4096 and 0.8192	24 and 72	3
Torre Pellice	24	2	1	2	46	6
Combined active and passive surface-wave tests						
La Salle	24 and 48	1.5 and 2	0.5	4.096	48 and 75	2
	12 and 24	10.35 and 19.41	16	524.288	Circular diameter 40 and 75	
Pianola	48	1	0.5	2.048	48	3
	12	12.94	8	524.288	Circular diameter 50	
Rojo Piano	48	1.5	0.5	2.048	70.5	3
	12	12.94	8	524.288	Circular diameter 50	
Catania	48	1.5	0.5	2.048	70.5	3
	12	12.94	8	524.288	Circular diameter 50	
Pisa	24	2.5	2	4.096	60	5
	8	15.31	2	688.128	Circular diameter 40	

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To evaluate the experimental uncertainty of both active-source and passive-source data sets, each record (shot or microtremor registration) has been independently processed. The experimental dispersion curve for the site is given by the mean velocity value at each frequency, and its related uncertainty is represented by the standard deviation. This procedure may lead to an overestimation of the uncertainty because no signal averaging is performed before calculating the standard deviation. However, on the other side, other sources of uncertainties (e.g., geometrical errors in the receiver array) are not taken into account. Altogether, a reasonable estimate is achieved, as confirmed also by the comparison to previous literature data obtained with different procedures (Marosi and Hiltunen 2004a, b).

A Monte Carlo procedure has been used for the inversion of experimental surface-wave dispersion curves (Socco and Boiero 2008). The algorithm uses scale properties of Rayleigh wave propagation (Socco and Strobbia 2004; Socco and Boiero 2008) to efficiently explore the model parameter space, cutting down the required number of forward simulations. The analyses for the present study were performed with a population of 10^5 profiles for each data set. The forward modeling algorithm is based on the Haskell (1953) and Thomson (1950) approach, and the fundamental mode solution has been considered. The misfit function compares the experimental and synthetic dispersion curves accounting also for experimental data uncertainty and problem dimensionality (number of unknown parameters with respect to available data points). It can be written as

$$\chi^2 = \frac{\sum_1^m (V_t - V_e)^2 \cdot \sigma_e^{-2}}{m - (2 \cdot n) - 1} \tag{1}$$

where V_t and V_e = theoretical and experimental phase velocities, respectively; σ_e contains the experimental data uncertainties; m = number of points in the dispersion curve; and n = number of layers in the model (only shear-wave velocity and thickness for each layer are considered unknown whereas the densities and Poisson ratios are fixed a priori). The misfit is evaluated for each profile of the population and used to select acceptable models according to a statistical test. The statistical one-tailed Fisher test (Sachs 1984) selects all profiles that are equivalent, given a confidence level. For the present study, a level of confidence equal to 1% has been used. The selected models hence represent a set of possible solutions that may be considered equally probable, given the experimental data and their uncertainty, the adopted model parameterization, and the level of confidence. Socco and Boiero (2008) showed that the solutions of deterministic inversions performed with different initial models fall in a wider model region with respect to the models selected by the Monte Carlo inversion. The profile set can therefore be assumed as a picture of the solution nonuniqueness for the specific data set. More details on the algorithm are reported in Socco and Boiero (2008).

Results of the Monte Carlo inversions are reported in all of the figures in this paper using a graphical scale from light to dark tones on the basis of the relative misfit of each shear-wave velocity profile. The darkest tone always corresponds to the model having the lowest misfit with reference to the experimental dispersion curve. The same tone is used to represent each shear-wave velocity model and its associated dispersion curve.

The $V_{S,30}$ has been computed for the population of equivalent shear-wave velocity profiles of each data set. This allowed retrieving the mean $V_{S,30}$ for each site and its associated uncertainty.

Case Histories

The locations of the sites in Italy are reported in Fig. 1. Different soil classes and soil conditions are investigated so that the data set presents a wide range of results in terms of $V_{S,30}$. Either by using heavy sources in active-source tests, or a combined interpretation of active-source and passive-source data, the experimental dispersion curve has been recovered for each site in a wide frequency band; this allows a $\lambda_{\max}$ greater than 60 m to be retrieved for all data sets. This is an essential requirement for $V_{S,30}$ determination. The statistical results of the Monte Carlo inversion for each site are summarized in Table 2.

The first site is located in La Salle (Valle d'Aosta) on a wide fluvial fan (Socco et al. 2008). The expected maximum thickness of the Quaternary deposits in La Salle is around 200 m, and the fan is mainly composed of fluvial deposits (sand, gravel, or stone), polygenic slivers, pebbles, and blocks. The experimental dispersion uncertainty in active-source and passive-source tests is considerably different as seen in the error bars reported in Fig. 2(a). The values of the CoV obtained from active data (approximately 1%) are in agreement with other studies (e.g., Marosi and Hiltunen 2004, b; Lai et al. 2005), whereas passive data, being based on natural seismic noise, show a higher variability (mean CoV around 4–6%). In the frequency band in which the two dispersion curve branches are superimposed (from approximately 15–18 Hz), there is, however, a good consistency of the experimental phase velocity values (see also Foti et al. 2007). The CoV increases with wavelengths because of CoV frequency dependence and effect of different kinds of sources. The same trend in CoV has been noticed for the other case histories in which both types of acquisitions (active and passive source) have been performed. Although the site was not heavily urbanized and far from major roads, a dominant source of microtremors was recognized, likely related to some excavation works in progress approximately a kilometer apart from the survey area (Foti et al. 2007); in this condition of localized source, passive linear arrays methods (such as ReMi) could fail in the determination of the correct phase velocity.

The 16 profiles selected by the Monte Carlo procedure are reported in Fig. 2(b). The results are compared to the shear-wave velocity profile obtained with a downhole test. The uncertainty in

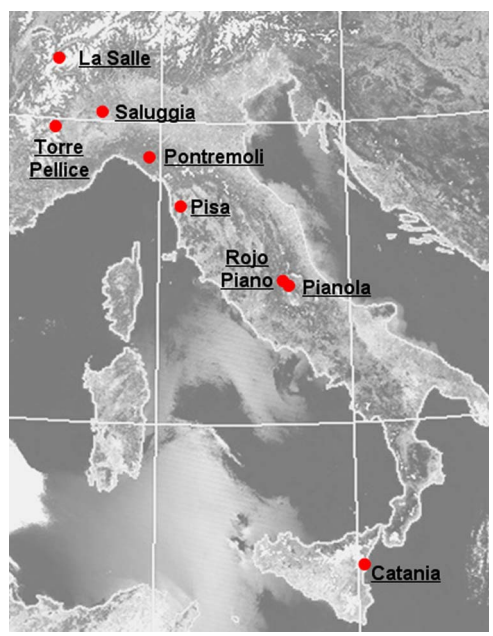

Fig. 1. Geographical location of test sites in Italy [These data are distributed by the Land Processes Distributed Active Archive Center (LP DAAC), located at the U.S. Geological Survey (USGS) Earth Resources Observation and Science (EROS) Center (lpdaac.usgs.gov)]

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Table 2. Summary of the Results of Monte Carlo Inversion of Surface-Wave Tests at the Italian Sites and Comparison with Invasive [Cross-Hole (CH) or Downhole (DH)] Test Results

	λ_{max} SW (m)	Number of equivalent profiles	Mean $V_{S,30}$ SW (m/s)	Standard deviation $V_{S,30}$ (m/s)	Mean CoV $V_{S,30}$ (%)	$V_{S,30}$ (m/s) invasive
Active-source surface-wave tests						
Pontremoli	63	22	774	15.2	2.0	797 (DH)
Saluggia	85	39	452	7.8	1.7	380 (CH)
Torre Pellice	145	16	319	7.4	2.1	294 (DH)
Combined active-source and passive-source surface-wave tests						
La Salle	155	16	491	8.2	1.7	491 (DH)
Rojo Piano	110	21	312	3.6	1.1	290 (DH)
Pianola	90	33	303	6.7	2.2	308 (DH)
Catania	110	24	162	3.5	2.1	195 (DH)
Pisa	90	25	181	2.0	1.1	176 (CH)

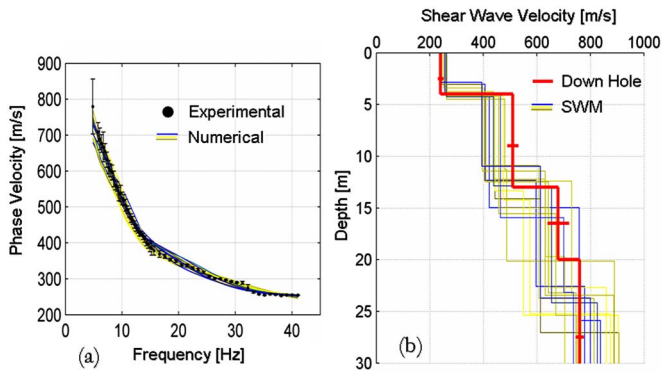

Fig. 2. Monte Carlo inversion of combined active and passive surface-wave data at La Salle site: (a) experimental and numerical dispersion curves; (b) shear-wave velocity profiles from Monte Carlo analysis compared to downhole test results

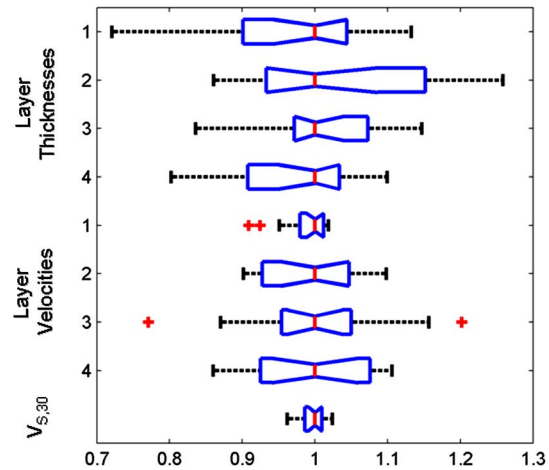

Fig. 3. Box plots of the distributions of model parameters and $V_{S,30}$ for the combined active and passive tests at La Salle site

the downhole test result is also available (Socco et al. 2008). The $V_{S,30}$ from the downhole test has a CoV of approximately 3%, which is comparable to the one obtained from surface-wave tests (Table 2).

Box plots of both model parameters (V_S and thicknesses) and of the final $V_{S,30}$ are reported in Fig. 3. Values are normalized with respect to the median for a comparable assessment of reliability. Box plots identify the statistical distribution of each parameter with an area delimited by the lower and upper quartiles (middle line is the median). The notches around the median value represent an estimate of the uncertainty for box-to-box comparison. The whisker lines, extending from each end of the boxes for 1.5 times the corresponding interquartile distance, show the extent of the rest of the data. Crosses are outliers with values beyond the ends of the whiskers. Shear-wave velocity of the shallowest layer presents the lowest uncertainty, as expected taking into account the resolution of surface-wave tests close to the ground surface and the large amount of information in the high frequency band, which makes the estimate very reliable. The uncertainty on the global $V_{S,30}$ is much lower than the uncertainty on each single model parameter. The reliability of a single model parameter is in fact strongly affected by solution nonuniqueness. This is a typical feature of the solution of inverse problems: several combinations of model parameters yield equally good profiles. As a consequence, a single parameter is not well resolved, but the overall model is reliable, as shown by the lower uncertainty on $V_{S,30}$. Similar conclusions have been obtained with respect to seismic site-amplification studies (Foti et al. 2009).

The second site is located in Torre Pellice (Socco et al. 2009); this case history is interesting because a velocity inversion is expected in the shallowest 30 m. Indeed, the geology of the site is characterized by a shallow formation of fluvial sediments with an expected variable thickness of 10–50 m over soft lacustrine sediments. The bedrock is expected to be more than 100 m deep in the central part of the valley whereas it is shallower on the lateral portions where tests have been executed. The results, reported in Fig. 4, confirm the presence of a velocity inversion relative to the contact between fluvial and lacustrine sediments (also confirmed by the downhole test performed nearby).

The third case history is located in Pontremoli and the local geology reports the presence of relatively shallow bedrock (Foti 2002). The sounding executed for the downhole test indicated the presence of a deposit of gravels and sands over the limestone bedrock located approximately at 10-m depth. Surface-wave and downhole test results are very similar at this site (accounting for the different parametrization) and the variability of profiles extracted from the Monte Carlo inversion is small (Fig. 5).

The Saluggia test site (Fig. 6) is located close to the Dora Baltea River, in a large flat area of fluvial sediments (Foti 2000). The soil is composed basically of gravels and gravelly sands, with the presence of fine sand and clayey silt, in the form of lenses. For this site, the profiles selected from the Monte Carlo inversion show a higher

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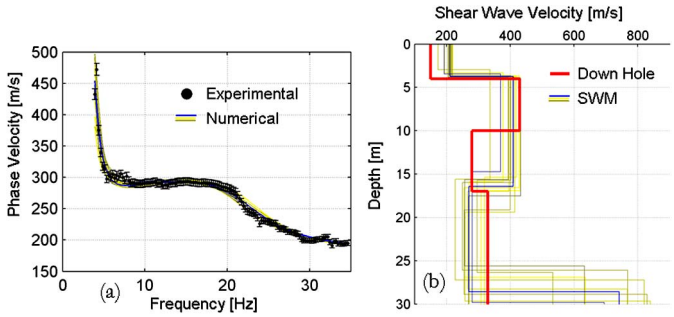

Fig. 4. Monte Carlo inversion of active surface-wave data at Torre Pellice site: (a) experimental and numerical dispersion curves; (b) shear-wave velocity profiles from Monte Carlo analysis compared to downhole test results

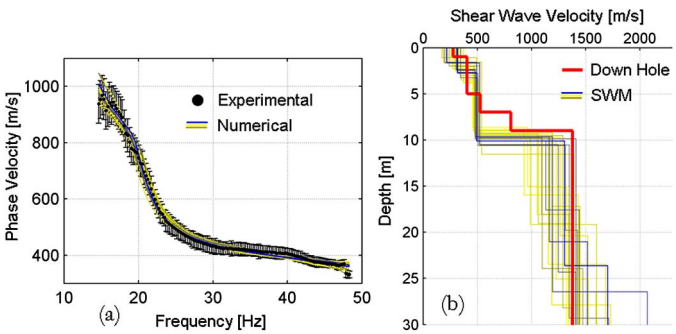

Fig. 5. Monte Carlo inversion of active surface-wave data at Pontremoli site: (a) experimental and numerical dispersion curves; (b) shear-wave velocity profiles from Monte Carlo analysis compared to downhole test results

velocity trend for depths below 15 m with respect to the cross-hole test. This reflects in a higher value of $V_{S,30}$ (452 m/s) than the one obtained with the cross-hole test (380 m/s). The Catania test site, on the contrary, shows a lower value of $V_{S,30}$ determined with surface-wave tests compared to the invasive test (Fig. 7). The Catania test site is in a wide sedimentary flat area mainly constituted by alluvial deposits (clays and silty clays) with some intercalated sands (Capilleri et al. 2009); The site is very close to the location of a station of the Italian Accelerometric Network. Results of the

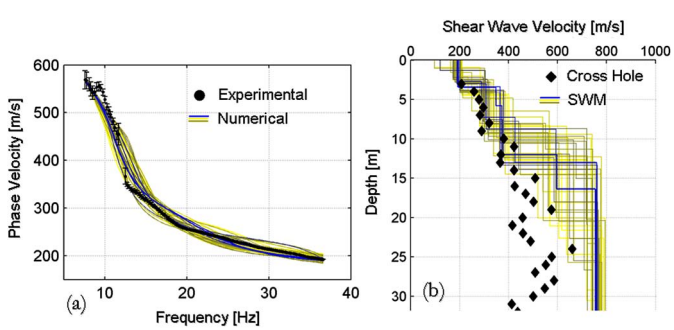

Fig. 6. Monte Carlo inversion of active surface-wave data at Saluggia site: (a) experimental and numerical dispersion curves; (b) shear-wave velocity profiles from Monte Carlo analysis compared to cross-hole test results

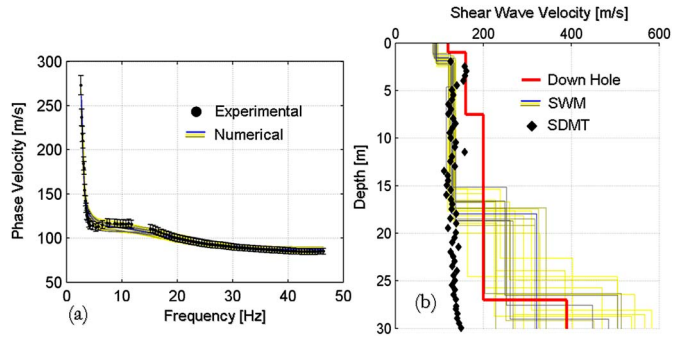

Fig. 7. Monte Carlo inversion of combined active and passive surface-wave data at Catania site: (a) experimental and numerical dispersion curves; (b) shear-wave velocity profiles from Monte Carlo analysis compared to downhole and SDMT test results

Monte Carlo inversion are reported in Fig. 7 and compared with the results of a downhole test at the same site (Capilleri et al. 2009) and of a seismic dilatometer test (SDMT) executed in a nearby site (D. Marchetti, personal communication, 2010). In fact the SDMT site is approximately 3.5 km away with also a few meters of difference in elevation, hence it cannot be considered for a strict comparison; however, the comparison can be considered meaningful for shallow layers because of the uniformity of the alluvial deposits along the whole Catania plain (Capilleri et al. 2009)

The Pisa site is located near the leaning tower and has been characterized within the remediation studies for the tower (Foti 2003). The well-characterized subsoil of this site is mainly constituted by clay formations with some slightly stiffer layers. The results of the Monte Carlo inversion (Fig. 8) identify a velocity inversion between 5 and 10 m that is more evident in the cross-hole test results.

Finally two sites near L'Aquila (Rojo Piano and Pianola) have been characterized within the activities for the selection of new construction sites in the aftermath of the 2009 earthquake. Both sites are located on the lacustrine deposits of the Aquilian Basin, which are mainly composed by fines and sands (Monaco et al. 2010). On the basis of available geological information, at both sites the bedrock is expected to be deeper than 30 m. For both sites, the results of a downhole test and of an SDMT are available for comparison (Monaco et al. 2010) and are reported in Figs. 9(b) and 10(b) together with the surface-wave inversion results.

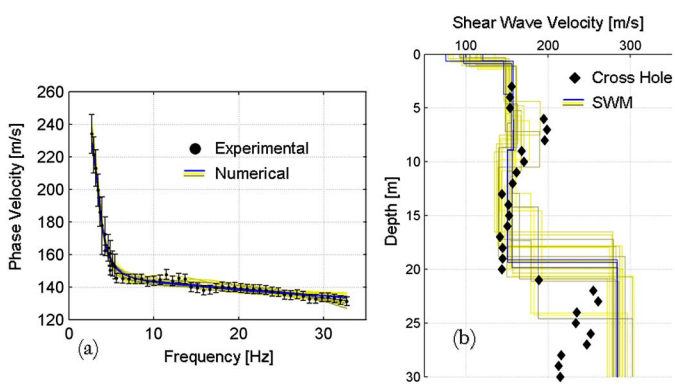

Fig. 8. Monte Carlo inversion of combined active and passive surface-wave data at Pisa site: (a) experimental and numerical dispersion curves; (b) shear-wave velocity profiles from Monte Carlo analysis compared to downhole test results

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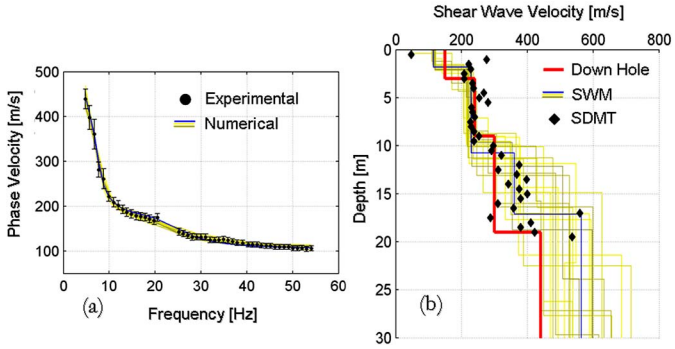

Fig. 9. Monte Carlo inversion of combined active and passive surface-wave data at Rojo Piano site: (a) experimental and numerical dispersion curves; (b) shear-wave velocity profiles from Monte Carlo analysis compared to downhole test results

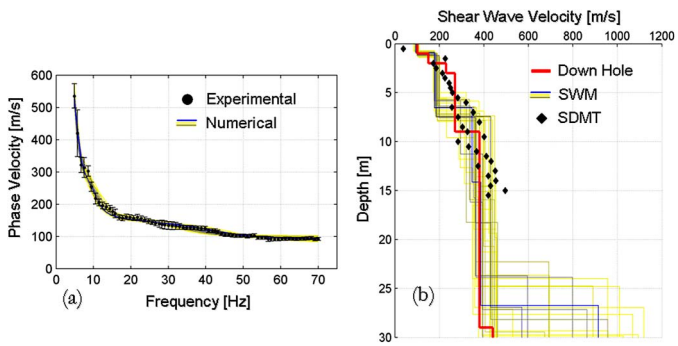

Fig. 10. Monte Carlo inversion of combined active and passive surface-wave data at Pianola site: (a) experimental and numerical dispersion curves; (b) shear-wave velocity profiles from Monte Carlo analysis compared to downhole test results

Discussion

For all of the sites, the uncertainty on $V_{S,30}$ (as quantified by the CoV) is more or less around 2% (Table 2) and lower than the measurement uncertainty of the experimental dispersion curves. This indicates that if the experimental dispersion curve is sampled in a sufficiently wide frequency band, the profiles selected by the Monte Carlo inversion are equivalent in terms of both fit to the dispersion curves and $V_{S,30}$. A similar equivalence has been demonstrated in Foti et al. (2009) in respect of both amplification functions and response spectra within the typical framework of one-dimensional (1D) seismic-site-response studies.

It is worthwhile to recall that different inversion strategies could lead to any of the profiles selected by the Monte Carlo procedure. The difference between velocity profiles can be significant, but in terms of $V_{S,30}$ evaluation this difference collapses to very limited values.

Comparison with Invasive Tests

A comparison of $V_{S,30}$ evaluation from invasive and surface-wave data is reported in Fig. 11 for the test sites presented in this study and for a series of data available in literature (Moss 2008, Fig. 3) concerning both spectral analysis of surface wave (SASW) and multichannel analysis of surface wave (MASW) methods. Although the comparison is generally good, Moss (2008) has underlined in his study that surface-wave tests tend to overestimate $V_{S,30}$ in soft sites and underestimate it in stiff sites. This trend, however, is not present

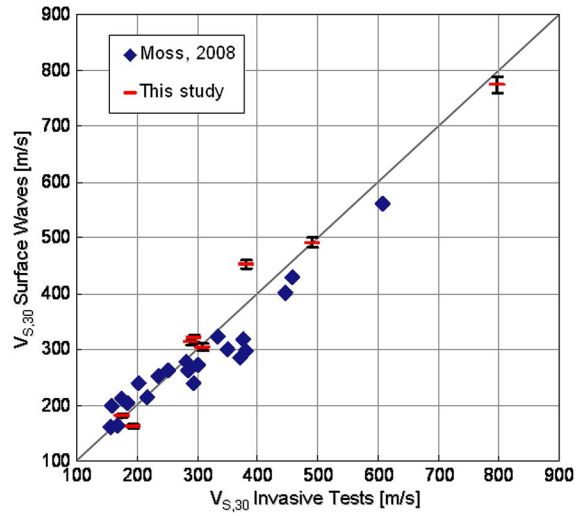

Fig. 11. Comparison of $V_{S,30}$ determined with invasive tests and surface-wave tests

in the data set of the present study in which the relationship appears closer to a 1:1 ratio. One of the possible explanations is given by the presence in the Moss data set of comparisons with P-S suspension logging, which tends to overestimate V_S in stiff materials. Data from this study shown in Fig. 11 are relevant as they extend the comparison of invasive versus surface-wave tests to stiffer soil conditions, even if much work is still to be done in this velocity range to confirm the observed trends. Moreover, the uncertainty bars associated with surface-wave tests show the limited relevance of solution nonuniqueness in this context.

Although the results from invasive tests are typically considered as a benchmark, there are several possible reasons for differences between surface-wave tests and borehole methods. Indeed, the investigated volume is different: surface-wave tests measure an average value based on the dynamic response of a large portion of the soil deposit, which can be considered more representative than local estimates obtained with invasive methods. It is also necessary to account for possible inaccuracies in borehole methods. For example, it is common experience that the downhole test become less accurate as a function of depth because of the increased source to receiver distance and that cross-hole tests can be biased because of the effects of drilling disturbance or because of critical refraction at interfaces. Furthermore, borehole measurements are strongly influenced by material anisotropy, borehole inclination and source polarization that are rarely addressed in their interpretation. Soil disturbance can produce either a decrease or an increase in velocity because of strain softening or hardening occurring around the borehole. This effect is particularly evident at softer sites. For example, data presented in Fig. 9 show that a less invasive methodology (SDMT) is more in agreement with the results of surface-wave tests than the downhole test. The same trend has been noted also at the Catania site (Fig. 7). Here it is necessary to recall that the SDMT was not located at the same site as the other two tests; still, a very good correspondence for shallow clays is obtained with surface-wave tests, whereas for higher depths (below 20 m) the difference is likely because of geological variations.

The cross-hole test at Saluggia was performed approximately 30 years ago with old style equipment and by using a 2-borehole setup; hence, the results could be affected by errors (e.g., inaccurate triggering or borehole inclination). This fact could justify the discrepancy with respect to surface-wave results [Fig. 6(b)]. On the other hand, it has to be observed that, among all the surface-wave

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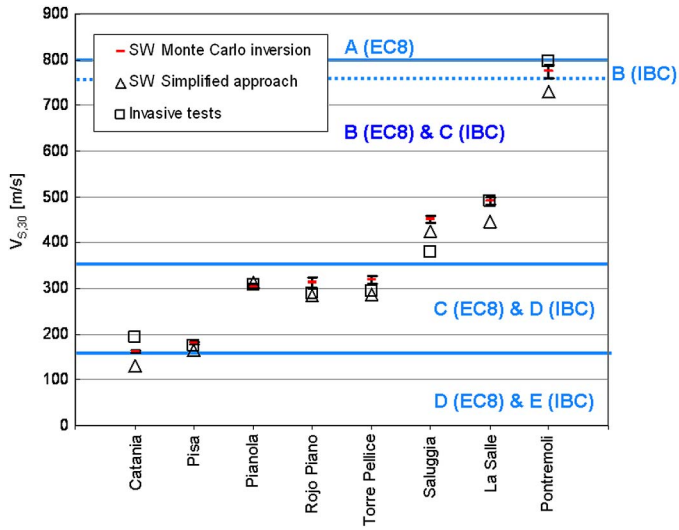

Fig. 12. Estimates of $V_{S,30}$ from surface-wave data using the Monte Carlo inversion and the simplified approach at different sites compared to estimates from invasive methods. Also the bounds for seismic site classification according to IBC (BSSC 1994) and EC8 (CEN 2004) are reported

data sets, this one shows the poorest fitting of the experimental dispersion curve in the inversion [Fig. 6(a)]; hence, in this case, the accuracy of the surface-wave test may also be questionable.

Simplified Procedure for $V_{S,30}$ Approximation

A preliminary estimate of $V_{S,30}$ can be obtained directly from the experimental dispersion curve. The idea that the dispersion curve offers a consistent and simplified way of estimating the $V_{S,30}$ has been considered by Brown et al. (2000), who proposed the Rayleigh-wave phase velocity at a wavelength of 36 m as a valuable estimate of $V_{S,30}$. Their analysis is on the basis of several case histories and the CoVs of their simplified method rises to around 6% for the different sites. Although such an approach cannot be considered as an alternative to the solution of the inversion process because this oversimplification can lead to large errors, especially for complex stratigraphic situations, it can provide a useful reference for preliminary assessments.

Results from the application of this simplified procedure for the Italian sites are compared to the outcome of the Monte Carlo inversion of surface-wave data and the results of invasive tests in Fig. 12. The sites have been sorted for increasing $V_{S,30}$. In the same figure the boundaries proposed by international building code (IBC) and Eurocode 8 (EC8) for site classification are reported for reference. The simplified procedure leads to reasonable results, giving a site classification that is the same as invasive tests and surface-wave inversion at most sites (neglecting the stiffest and softest sites, which are borderlines). The root-mean-square deviation of this simplified procedure with respect to invasive tests is around 7%.

Final Remarks

This paper has focused on the reliability of surface-wave tests for $V_{S,30}$ determination. One of the frequent criticisms of surface-wave methods (SWMs) is related to the nonunique solution of the inverse problem. It has been shown, however, that this nonuniqueness has a reduced influence in the evaluation of $V_{S,30}$. The $V_{S,30}$ is related to a global response of the soil profile, so that differences in estimated profile parameters, which are associated with small differences in the dispersion curve (equivalence problem) are of minor importance for its estimate. Indeed, even if different inversion approaches can lead to different solutions in terms of a shear-wave velocity profile, the associated estimate of $V_{S,30}$ is very similar (i.e., within approximately 2%) if the experimental data cover the necessary frequency band. This confirms the robustness of surface-wave testing for estimating $V_{S,30}$. With respect to the accuracy, the reported case histories show that, in situations in which the maximum retrieved wavelength (either with active or passive sources) is greater than 60 m, the final result is approximately close to the one of invasive tests. Therefore it appears that, in this condition, inversion nonuniqueness will not significantly alter the accuracy and reliability of the $V_{S,30}$ estimate. A single deterministic profile can yield reliable results in terms of $V_{S,30}$, if a good fitting of experimental dispersion curve is achieved.

These considerations are confirmed both in the sites of the presented data set and in previous literature data. With respect to the latter, the present work increases the coverage on stiff sites and provides a statistical evaluation of the uncertainty related to nonuniqueness of the solution.

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